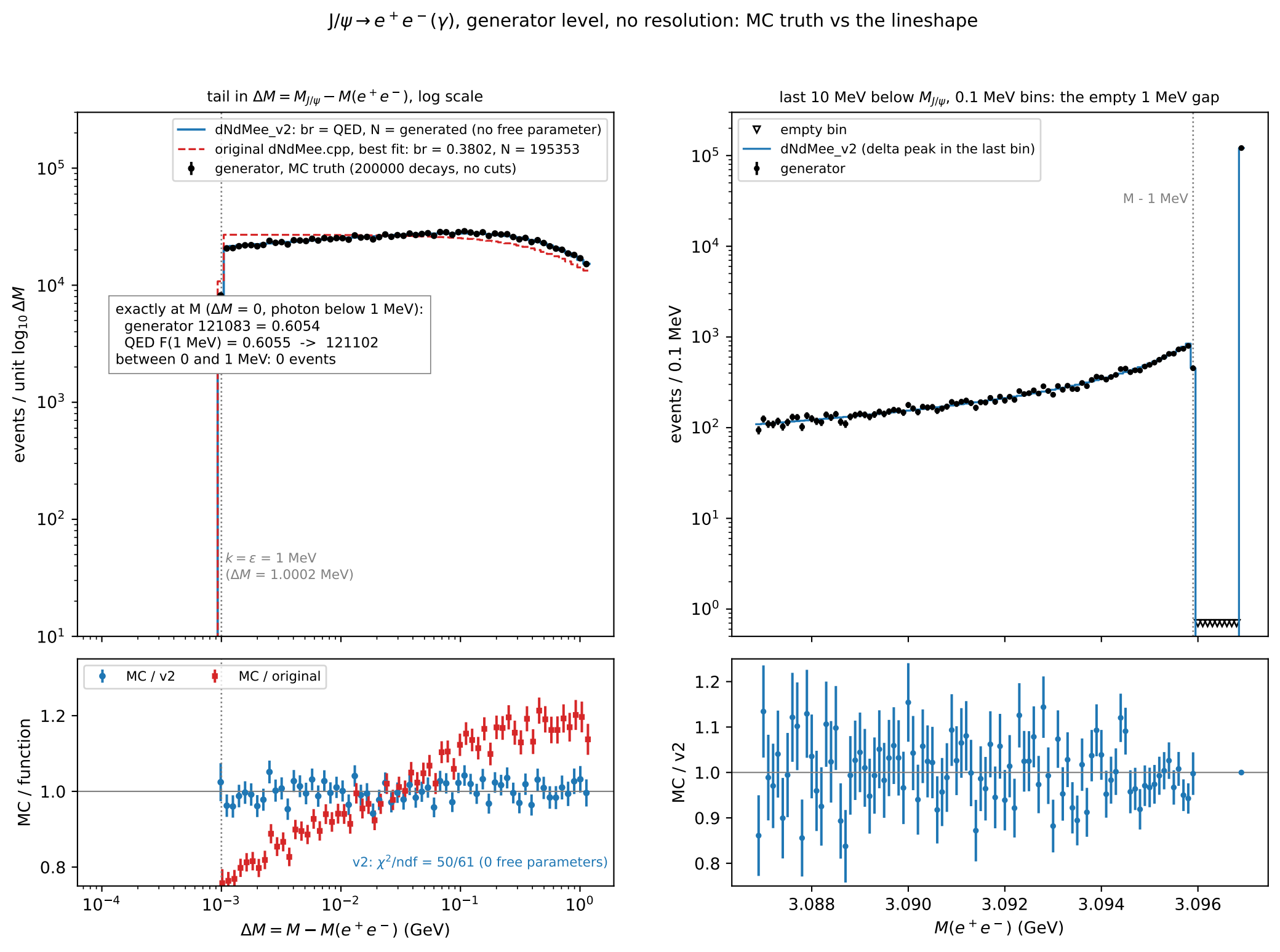
$J/\psi \to e^+e^-(\gamma)$, generator level, no resolution: MC truth vs the lineshape
tail in $\Delta M = M_{J/\psi} - M(e^+e^-)$, log scale
dNdMee_v2: br = QED, N = generated (no free parameter)
original dNdMee.cpp, best fit: br = 0.3802, N = 195353
generator, MC truth (200000 decays, no cuts)
exactly at M (ΔM = 0, photon below 1 MeV):
generator 121083 = 0.6054
QED F(1 MeV) = 0.6055 -> 121102
between 0 and 1 MeV: 0 events
$k = \varepsilon$ = 1 MeV
(ΔM = 1.0002 MeV)
events / unit $\log_{10}\Delta M$
MC / v2
MC / original
MC / function
v2: χ^2/ndf = 50/61 (0 free parameters)
$\Delta M = M - M(e^+e^-)$ (GeV)
last 10 MeV below $M_{J/\psi}$, 0.1 MeV bins: the empty 1 MeV gap
empty bin
dNdMee_v2 (delta peak in the last bin)
generator
M - 1 MeV
events / 0.1 MeV
MC / v2
$M(e^+e^-)$ (GeV)